CS 486/686
Supervised Learning & Generalization

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Lecture 17

RN 19.1–19.2, 19.6 · PM 7.1–7.3

Recorded lecture (asynchronous)

> 🎥 This lecture is **pre-recorded** — I'm away at **ICML** this week, so there's no in-person class today. Watch at your own pace.
>
> 📅 **Due today (Tue Jul 7):** Chat 8 and the CS686 project proposal.
>
> 💬 Questions? Post on **Piazza** — the TAs are available, and I'll follow up when I'm back.

Learning goals

- Name the **types of learning** and tell **classification** from **regression**.
- See supervised learning as **fitting parameters to minimize a loss**.
- Derive and fit **linear regression** and **logistic regression**.
- Use **cross-entropy** and **softmax** for classification.
- Explain **overfitting**; control it with **cross-validation** and **regularization**.

Why an agent that learns?

Medical diagnosis

Spam filtering

Chatbots (LLMs)

Speech

Image generation

Learning = improving on future tasks based on experience (data).

- We can't anticipate every situation, and the world keeps changing.
- For most of these tasks we have *no idea how to program a solution* — only how to *show examples*.

Three kinds of learning

Supervised

Given inputs + **targets**, predict targets for new inputs.

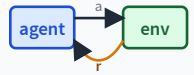
Today's focus.

Unsupervised

No targets — find **structure**: clustering, representations.

L18.

Reinforcement



Learn from rewards — what to *do*.

L15 (done).

Supervised or unsupervised?

Q1. We have a user's credit-card transactions and want to flag any that look *different* from the rest. We have no fraud labels.

A. Supervised

B. **Unsupervised**

> **B — Unsupervised.** No target labels to predict.

Q2. We have historical weather labels (sunny/cloudy/rain/snow) for a date, and we want to predict the same date next year.

A. **Supervised**

B. Unsupervised

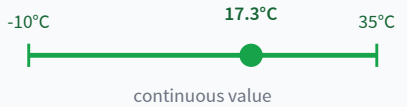
> **A — Supervised.** Each example has a target (the weather label).

Two flavors of supervised learning

Classification

Target is **discrete** — e.g. dog vs cat.

Regression



Target is **continuous** — e.g. tomorrow's temperature.

Q3. Predict next year's weather label (sunny/rain/...) for a date.

> **Classification** — discrete target.

Q4. Predict next month's price of a house.

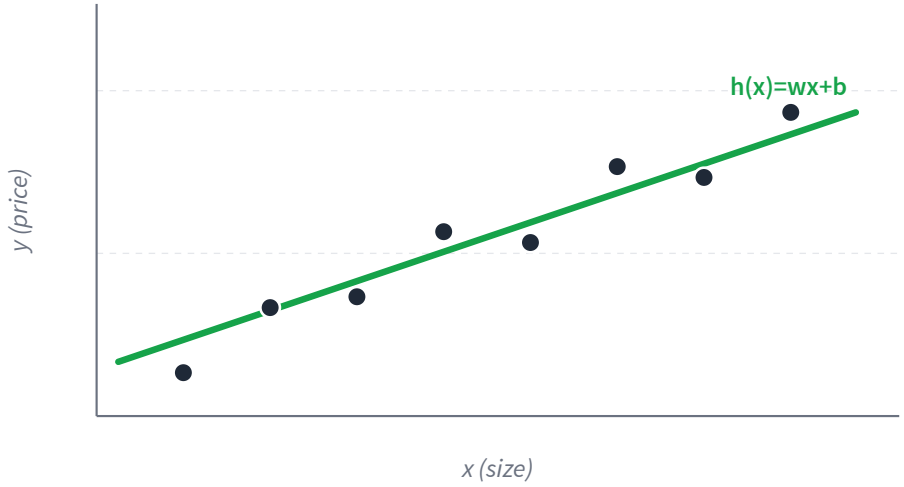
> **Regression** — continuous target.

The recipe behind every supervised learner

1. **Model.** Pick a function $h_{\mathbf{w}}$ with tunable **parameters** $\mathbf{w}$ (weights).
2. **Loss.** Measure how wrong $h_{\mathbf{w}}$ is: $L(\mathbf{w}) = \frac{1}{m} \sum_i \ell\big(h_{\mathbf{w}}(x_i),\, y_i\big)$.
3. **Fit.** Choose $\mathbf{w}$ that makes $L(\mathbf{w})$ small — usually by **gradient descent**.

Change the model and the loss → you get every method in this module (regression → neural nets → transformers).

Linear regression: the model



One real input x, one real output:

$$h_{\mathbf{w}}(x) = w\,x + b$$

With n features $\mathbf{x} = (x_1, \ldots, x_n)$:

$$h_{\mathbf{w}}(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$$

Parameters to learn: the slope(s) $\mathbf{w}$ and intercept b.

Linear regression: fitting by least squares

Loss = mean **squared error** between prediction and target:

$$L(\mathbf{w}, b) = \frac{1}{m} \sum_{i=1}^{m} \big(h_{\mathbf{w}}(x_i) - y_i \big)^2$$

Closed form

L is convex — set $\nabla L = 0$ and solve. For features X: $\mathbf{w}^\star = (X^\top X)^{-1} X^\top \mathbf{y}$.

Gradient descent

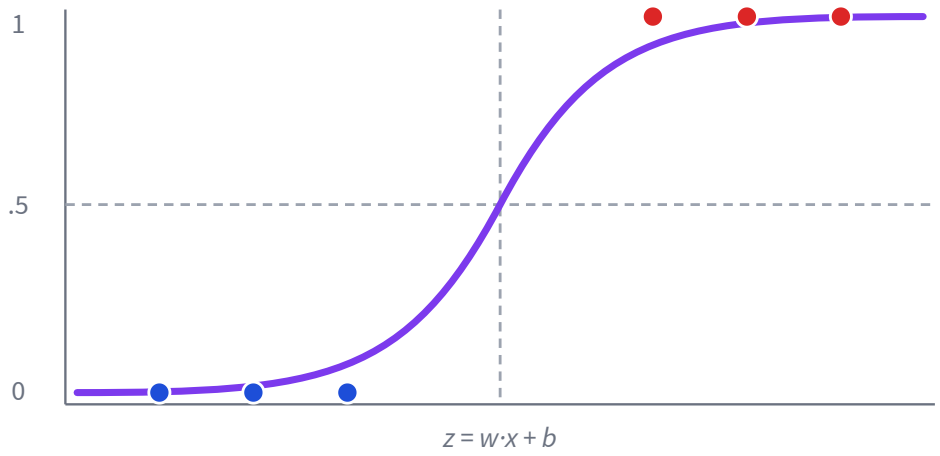
Works for *any* differentiable model.

Repeat:

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \, \nabla_{\mathbf{w}} L$$

Gradient descent is the workhorse for the rest of the module — the models just get bigger.

Classification: logistic regression



Squash the linear score into a **probability** with the sigmoid:

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}, \; z = \mathbf{w}^\top \mathbf{x} + b$$

- $\hat{y} = P(y{=}1 \mid \mathbf{x})$.
- Predict class 1 when $\hat{y} > 0.5$, i.e. $z > 0$.
- The boundary $z = 0$ is a **hyperplane**.

Loss for classification: cross-entropy

Squared error is a poor fit for probabilities. Instead, maximize the probability of the true labels — equivalently, minimize **binary cross-entropy**:

$$L(\mathbf{w}) = -\frac{1}{m}\sum_{i=1}^{m}\Big[\, y_i \log \hat{y}_i + (1 - y_i)\log(1 - \hat{y}_i)\,\Big]$$

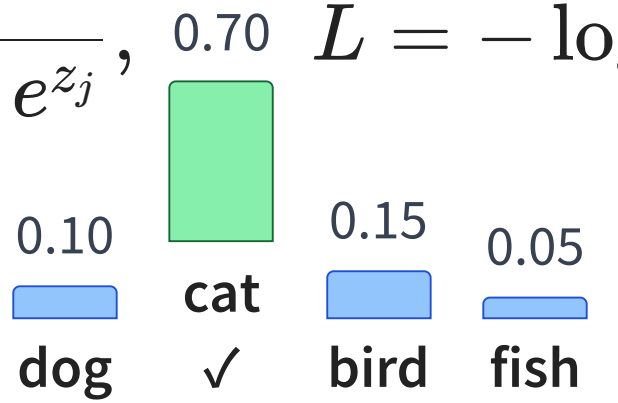
True label $y = 1$: loss is $-\log \hat{y}$ — small when $\hat{y} \to 1$, **huge** when $\hat{y} \to 0$.

Confident *and* wrong is punished the hardest — exactly the behaviour we want.

Many classes: softmax + cross-entropy

Output one score z_k per class; **softmax** turns scores into a probability distribution:

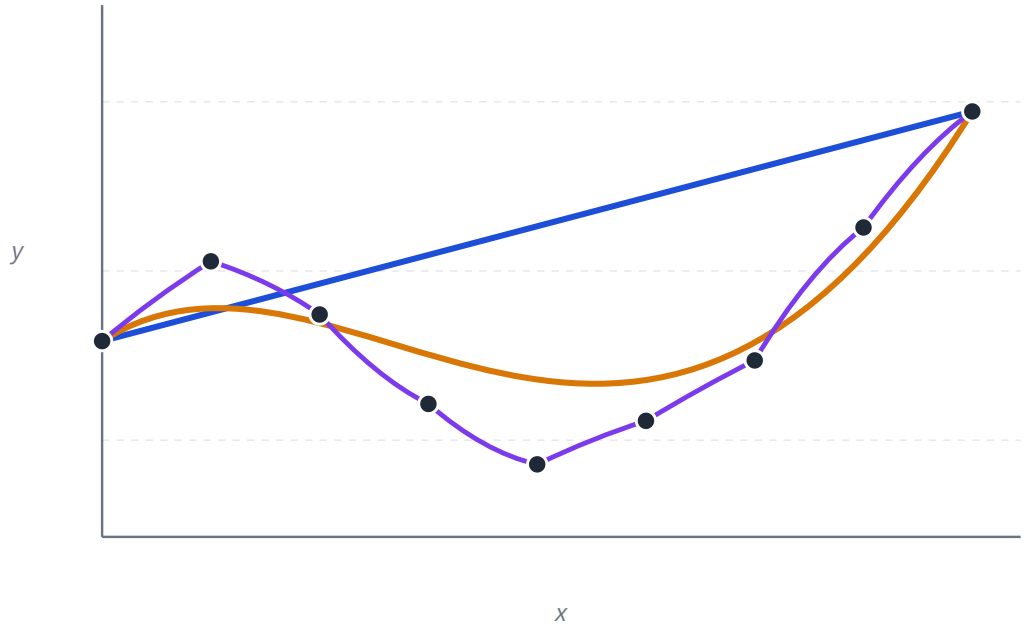
$$P(y{=}k \mid \mathbf{x}) = \frac{e^{z_k}}{\sum_j e^{z_j}}, \quad L = -\log P(y{=}\text{true class})$$



This softmax + cross-entropy is exactly the output layer we'll reuse for neural nets (L20).

More power is not always better

Same 9 points, increasingly flexible models. The wiggliest one fits training data best — but would you trust it on a new x?



degree 1 (underfits) degree 3 (good fit) degree 9 (overfits)

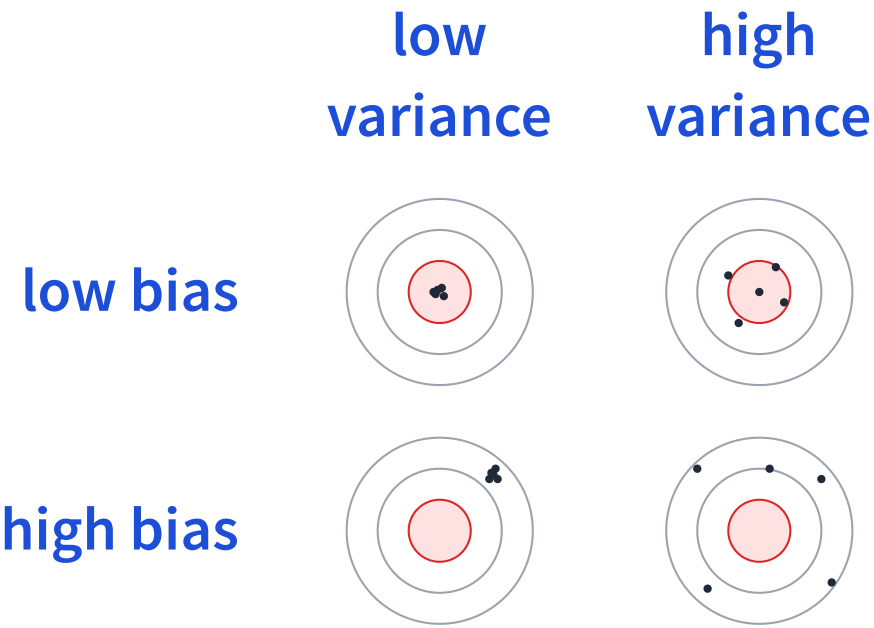
Bias and variance

Bias

Even with infinite data, how far off is the model family? High bias = too simple, **underfits**.

Variance

How much does the fit swing with a different training set? High variance = too flexible, **overfits**.



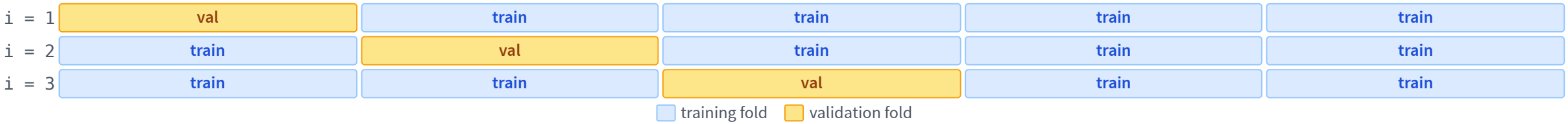
Dots = predictions from re-trained models.
Bullseye = truth.

K-fold cross-validation

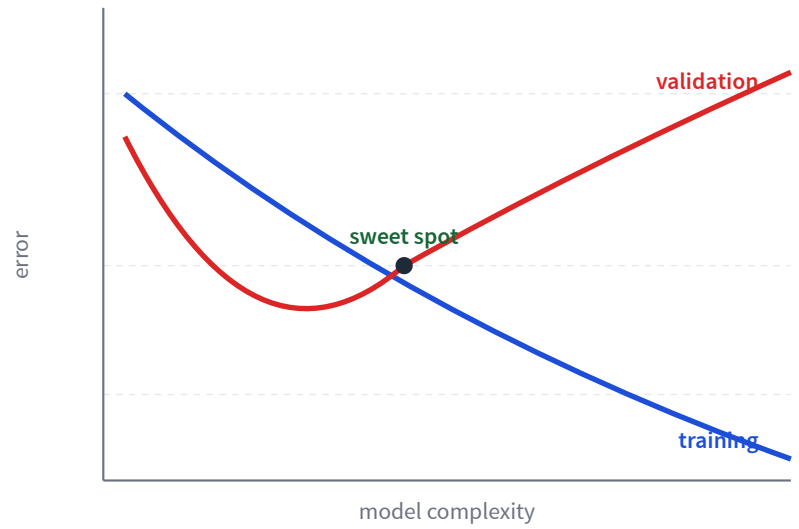
How to pick the right complexity? Use a slice of training data as a **surrogate test set**.

`k-fold-CV(model, dataset,` K`)`

1. Split training data into K equally sized folds.
2. For $i = 1, \ldots, K$: train on the other $K - 1$ folds; evaluate on fold i.
3. Average the error over the K runs; keep the hyperparameters with the best average.



Diagnosing & fighting overfitting



Training error keeps dropping; **validation** error turns back up — that gap is overfitting.

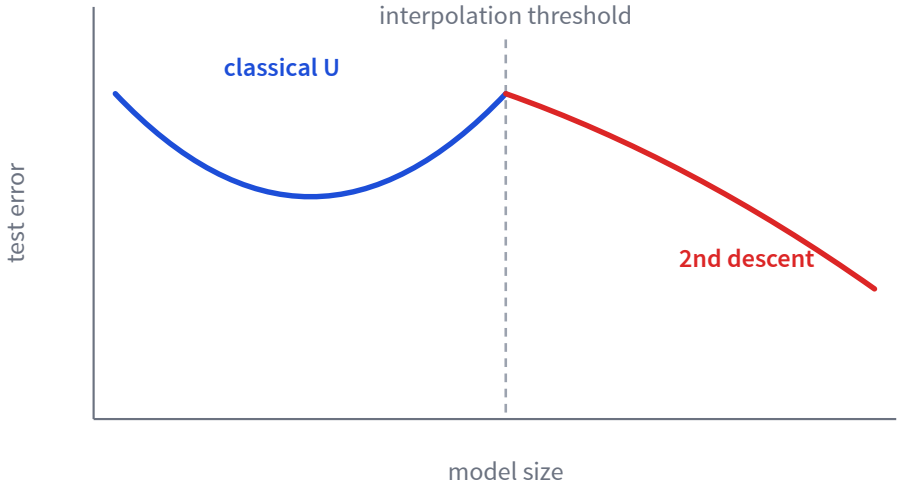
Regularization

Add a penalty on large weights to the loss:

$$L(\mathbf{w}) + \lambda \|\mathbf{w}\|^2$$

λ trades bias against variance; tune it with cross-validation.

A modern twist: double descent



The classical U-curve isn't the whole story.

- Past the point where the model fits training data *exactly*, test error can **fall again**.
- Very large models (deep nets, LLMs) often generalize *better*, not worse.
- Why? Still an active research question — "bigger" changed the rules.

Learning goals (recap)

✓ Name the types of learning; classification vs regression.

✓ Supervised learning = **model + loss + fit**.

✓ **Linear** and **logistic regression**; least squares & gradient descent.

✓ **Cross-entropy** and **softmax** for classification.

✓ Control overfitting with **cross-validation** and **regularization**.

Next: unsupervised & representation learning

Today, labels guided every hypothesis. Without labels, what *structure* can we still find in data?

L18: clustering (k-means), PCA, autoencoders, and learned representations.